PART-1 (20 Marks)

Note: Choose the best option and record your choice on the **Answer Sheet** following this part. If answers are not transferred to the sheet, they may not be graded.

1. If for a graph search problem, the cost of every action is the same, breadth first search is guaranteed to return an optimal solution.
(A) True (B) False

2. For a 3×3 tic-tac-toe game, there are __91__ different paths (from start to the end of game) in its state space graph.
(A) 3^9 (B) 9^3 (C) 8! (D) None of the given options

3. If symmetry of initial symbol placement is considered, the above tic-tac-toe game has ______ paths through its state space graph.
(A) $3 \times 7!$ (B) $12 \times 7!$ (C) $8 \times 7!$ (D) None of the given options

4. Which of the following is true about data-driven search?
(A) It is also called backward chaining (B) It starts at the goal node(s)
(C) It involves searching from facts to the goal(s) (D) It is feasible when goal(s) of a problem are known.

5. For a shallower search tree, depth first iterative deepening search compared with depth first search, in general, is:
(A) More efficient (B) Equally well (C) Optimal (D) Relatively inefficient

6. For the 8-puzzle problem, the state space has ______ states accessible from any random start state.
(A) 8! (B) 9! (C) $9!/2$ (D) None of the given options

7. $\alpha\beta$ search is generally better than the minimax search.
(A) True (B) False

8. β cut-off means:
(A) Prune further possibilities below the current MAX node
(B) Prune further possibilities below the current MIN node

9. If d represents depth and b, the average branching factor, the worst-case space complexity of $\alpha\beta$ search can be characterized as:
(A) $O(b^d)$ (B) $O(b \log d)$ (C) $O(d^b)$ (D) $O(bd)$

10. Admissibility of a heuristic always implies:
(A) Informedness (B) Monotonicity (C) Irrevocability (D) None of the given options

11. In the best first search, which of the following is true:
(A) A set of sorted paths is added to the queue
(B) Entire queue is sorted each time a new path is added
(C) Chronological backtracking is done to switch to promising paths
(D) All the nodes with a specified heuristic value(s) are explored first

12. _________ is a conceptual representation, while ___________ is a concrete one.
(A) Search Tree, State Space (B) State Space, Search Tree

13. Admissibility is always a less strict requirement than monotonicity.
(A) True (B) False

14. For certain problems, e.g. fractional knapsack problem, hill climbing is always guaranteed to give optimal solution.
(A) True (B) False

Scanned with CamScanner

15. Heuristic evaluation function for the uniform cost search may be characterized as:
(A) $f(n) = g(n) + h(n)$; where $h(n) = 0$ (C) $f(n) = g(n) + h(n)$; where both $g(n)$ and $h(n)$ are non-negative
(B) $f(n) = g(n) + h(n)$; where $g(n) = 0$ (D) None of the given options

16. Considering selection strategies in the genetic algorithm, roulette wheel selection enables more _______ than elitism.
(A) Exploitation (B) Exploration (C) None of the given options

17. TSP optimization problem is an NP hard problem.
(A) True (B) False

18. A smaller tournament size would result in a less diverse offspring population.
(A) True (B) False

19. Genetic algorithm can be characterized as a powerful variant of_______
(A) Uniform cost search (B) Breadth first search (C) Hill climbing search (D) None of the given options

20. Mutation helps ensure that _______ is present in the offspring population.
(A) Stagnation (B) Diversity (C) Inversion (D) None of the given options

Scanned with CamScanner

Answer Sheet

Question	Choice
1	A
2	D
3	A
4	C
5	D
6	C
7	A
8	~~B~~→A
9	D
10	D
11	B
12	B
13	A
14	A
15	A
16	A
17	A
18	A
19	C
20	B

17

Scanned with CamScanner

Draw the state

PART-2 (50 Marks)

Note: Attempt all questions.

Q.1. [1+2+4+3] 09

A farmer wants to move himself, a fox, a goose and some grain across a river. Unfortunately, his boat is so tiny that he can take only one of his possessions across the river on any trip and he is the one who could operate the boat. Worse yet, an unattended fox will eat a goose, and an unattended goose will eat grain! So, the farmer must not leave the fox alone with the goose or the goose alone with the grain. If the farmer and his possessions are all on the left bank of the river, what sequence of moves must the farmer follow to get to the right bank?

Consider representing the above problem using the state space representation as follows.

Using the 4-tuple representation for a problem state as $< a_1, a_2, a_3, a_4 >$, where the indices 1 through 4 represent the farmer, the fox, the goose, and the grain, respectively, and $a_i \in \{L, R\}$, where L and R designate left and right banks of the river, e.g. $< L, R, L, L >$, do the following.

(a) What would the start and goal states look like?

start ~~goal~~ = $<L, L, L, L>$ $a_i \in \{L, R\}$

Goal state = $<R, R, R, R>$

1

(b) Give the states that must be avoided given the constraints of the problem.

Avoided States:

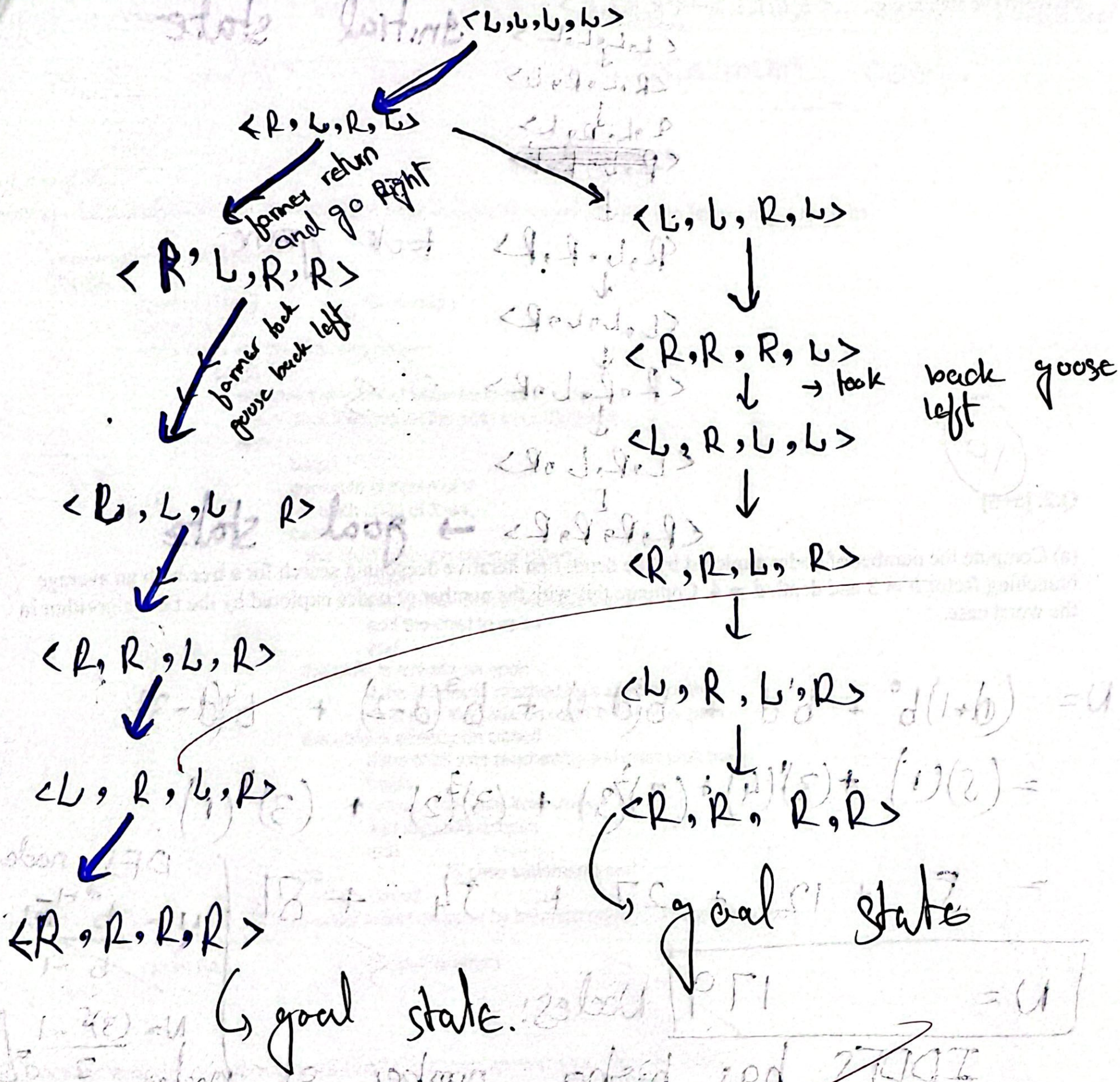
$<L, R, R, L>$
$<L, L, R, R>$

$<R, L, L, R>$
$<R, R, L, L>$

$<L, R, R, R>$
$<R, L, L, L>$

2

c) Draw the state space graph showing only the legal moves.



d) Give the sequence of moves to reach the goal state. Use the tuple notation for a state and arrows to show transi[tions] between the states, e.g., …< L, R, L, L >→< R, R, L, R > …

$\langle L, L, L, L \rangle$ Initial state

$\downarrow$

$\langle R, L, R, L \rangle$

$\downarrow$

$\langle L, L, R, L \rangle$

$\downarrow$

$\langle R, L, R, R \rangle$ took grane

$\downarrow$

$\langle L, L, L, R \rangle$

$\downarrow$

$\langle R, R, L, R \rangle$

$\downarrow$

$\langle L, R, L, R \rangle$

$\downarrow$

$\langle R, R, R, R \rangle$ → goal state

3

10

Q.2. [5+5]

(a) Compute the number of nodes explored by the depth first iterative deepening search for a tree with an average branching factor $b = 3$ and depth $d = 4$. Compare this with the number of nodes explored by the DFS algorithm in the worst case.

$$N = (d+1)b^0 + b^1 d + b^2(d-1) + b^3(d-2) + b^4(d-3)$$

$$= (5)(1) + (3)^1(4) + (3)^2(3) + (3)^3(2) + (3)^4(1)$$

$$= 5 + 12 + 27 + 54 + 81$$

$$\boxed{N = 179} \text{ Nodes!}$$

5

DFS nodes

$$N = \frac{b^{d+1} - 1}{b - 1}$$

$$N = \frac{(3)^5 - 1}{2} = \boxed{121}$$

IDDFS has higher number of nodes.

(b) Define admissibility of a heuristic function. What benefits does an admissible heuristic bring to the state space search?

Admissibility: A heurestic is the admissible if it never overestimates the true cost to reach the goal state from that state. It is true for all the nodes in that search space.

It's prove is given as: Is 'n'' the goal state?

$$0 \le h(n) \le h(n')$$ where $h(n')$ is the true cost and $h(n)$ is the estimated heuristic cost to reach goal.

Scanned with CamScanner

Admissibility ensures the optimal solution. If we reach at a certain ~~space~~ state, admissibility will ensure that we have reached there with the true minimum cost.

08

Q.3. [2+4+4]
Consider the pseudo code for the best first search algorithm and answer the following questions.

```
function best_first_search;
begin
        open := [Start]                    % initialize
        closed := [ ]
        while open ≠[ ] do, % states remain
                begin
                  remove the leftmost state from open, call it X
                  if X = goal then return the path from Start to X
                  else
                        begin
                        generate children of X
                        for each child of X do
                        case
                          the child is not on open or closed:
                                begin
                                assign the child a heuristic value
                                add the child to open
                                end
                          the child is already on open
                                if the child was reached by a shorter path
                                then give the state on open the shorter path
                          the child is already on closed
                                if the child was reached by a shorter path then
                                begin
                                remove the state from closed
                                add the child to open
                                end
                        end                % case statements end
                        put X on closed
                        re-order states on open by heuristic merit     % (best leftmost)
                end
        return FAIL                        % open is empty
end
```

(a) Does the pseudo code above work fine? Give your answer as a 'Yes' or 'No'.

'Yes' It'll work fine. 2

(b) Change the pseudo code if you answer No' to part (a) above. Note: Highlight the section you think is not correct and give the modified statement(s) below.

As the above psuedo code will work fine, so there is no need to change anything in the code. So the code will remain same.

Scanned with CamScanner

(c) **Modify the given pseudo code if the heuristic you use is monotone.**

As we're mentioning that heuristic we'll use is a monotone so it'll ensure that if a state is in the closed list, so it will already ~~be~~ reach by the shortest path (no need to update heuristic).
So the snapshot of code we will remove from the code to work it fine.

"the child is already on open
if child was reached by shorter path
then give state on open shorter path.
"

02·

Remove this part from code and it'll work for the heuristic we use is monotone.

Q.4. [4+6] 05

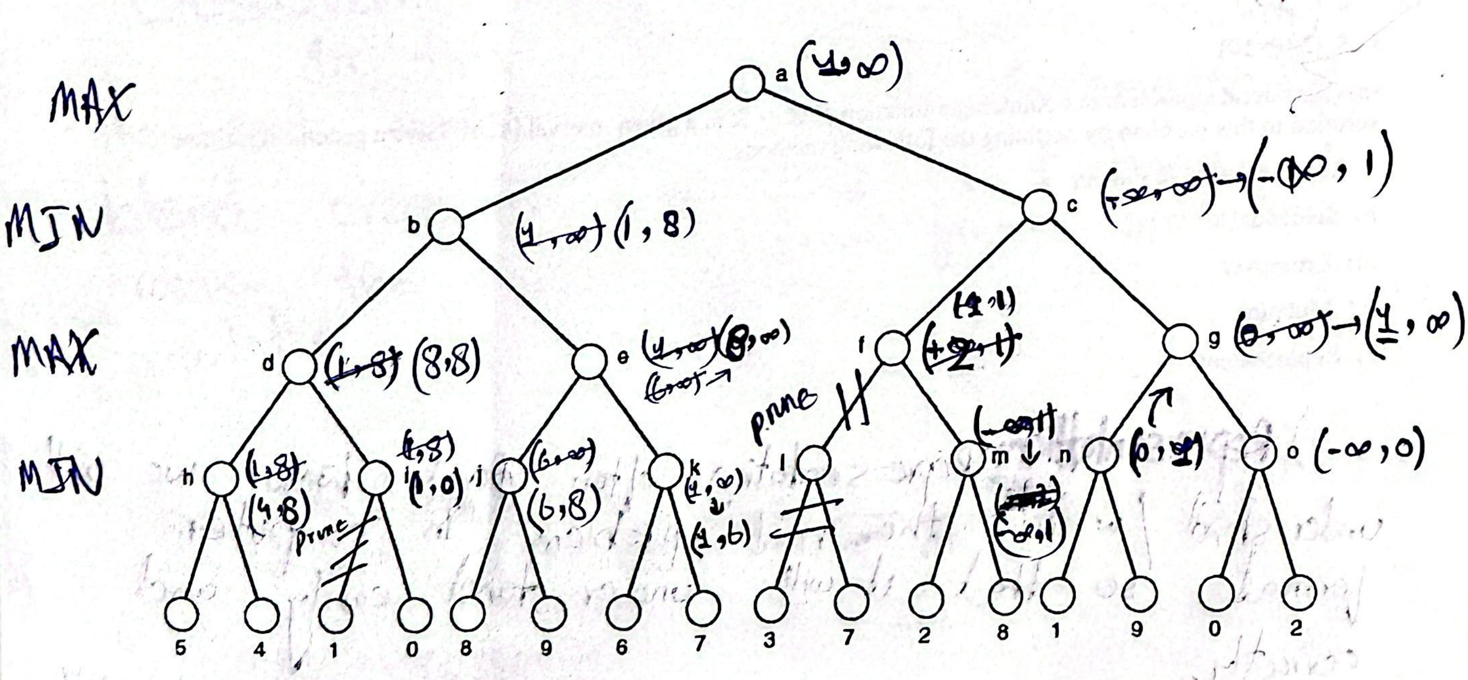
(a) What could be a problem with bounded lookahead in Minimax algorithm? Can $\alpha\beta$-search solve this?

Minimax algorithm traverse the nodes that we do'not need to look after where ($\alpha \geq \beta$) because they lead to more worst values of (α, β). Minimax algorithm travers every node of game tree which leads time complexity $O(b^d)$. $\alpha\beta$-search involves pruning where the $\alpha \geq \beta$ condition satisfies. Because traversing the node where $\alpha \geq \beta$ will not affect the overall ~~bound~~ search space. So its time complexity is $O(b^{d/2})$

02

(b) For the search tree shown below it is MAX's turn, i.e. node 'a' is MAX's node. You are required to compute the final values of the pair (α, β) for all non-leaf nodes generated by the algorithm and fill them in the given table. What move does the MAX select?

Note: Do the **right to left** depth-wise traversal. You must NOT evaluate (α, β) at the leaf nodes, which implies that the propagation of (α, β) pair must be stopped one level above the ply depth.



Scanned with CamScanner

Node	Final (α, β)
a	$(4, \infty)$
b	$(1, 8)$
c	$(-\infty, 1)$
d	$(8, 8)$
e	$(8, \infty)$
f	$(4, 4)$
g	$(4, \infty)$
h	$(4, 8)$
i	$(1, 0)$
j	$(6, 8)$
k	$(1, 6)$
l	$(-\infty, +\infty)$
m	$(-\infty, 4)$
n	$(0, 1)$
o	$(-\infty, 0)$

03

10

Q.5. [2×5=10]

(a) Consider the problem of optimizing a function $f: \mathbb{R} \to \mathbb{R}$ in a given interval $[a, b]$. Give a genetic algorithm solution to this problem by outlining the following methods:

i). Representation & Fitness

ii). Selection

iii). Crossover

iv). Mutation

v). Replacement

i) Representation: Representation often ~~Reads~~ how we will under stand / write the reed problem in computer format so that it will under stand easily and correctly.

Fitness: Fitness function evaluate how much a individual is close to the ~~op~~ global maximum.

for $f: \mathbb{R} \to \mathbb{R}$ in given interval $[a, b]$

Scanned with CamScanner